



Estd: 1986

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Sri Adichunchanagiri Shikshana Trust ®

SJC INSTITUTE OF TECHNOLOGY

Chickballapur – 562 101

Department of Artificial Intelligence and Data Science Tutorial - 3

SUBJECT TITLE	Database Management System
SUBJECT TYPE	Integrated
SUBJECT CODE	BCS403
ACADEMIC YEAR	2023-24
SCHEME	NEP CBCS 2022
SEMESTER	4 th sem, AI&DS
FACULTY NAME and DESIGNATION	Darshan A, Assistant Professor
DATE of ISSUED	26-07-2024

Q. No	Questions	Level	COs																													
1	EMPLOYEE <table><tr><td>Fname</td><td>Minit</td><td>Lname</td><td>Ssn</td><td>Bdate</td><td>Address</td><td>Sex</td><td>Salary</td><td>Super_ssn</td><td>Dno</td></tr></table> DEPARTMENT <table><tr><td>Dname</td><td>Dnumber</td><td>Mgr_ssn</td><td>Mgr_start_date</td></tr></table> DEPT_LOCATIONS <table><tr><td>Dnumber</td><td>Dlocation</td></tr></table> PROJECT <table><tr><td>Pname</td><td>Pnumber</td><td>Plocation</td><td>Dnum</td></tr></table> WORKS_ON <table><tr><td>Essn</td><td>Pno</td><td>Hours</td></tr></table> DEPENDENT <table><tr><td>Essn</td><td>Dependent_name</td><td>Sex</td><td>Bdate</td><td>Relationship</td></tr></table> a) Write a RA expression retrieve the name and address of all employees who work for the ‘Research’ department. b) Write a RA expression for every project located in ‘staffor’, list the project number, the controlling department number, and the department managers last name, address and birth date. c) Wrtie a RA expression to find the names of employees who work on all the projects controlled by departmnet number 5.	Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno	Dname	Dnumber	Mgr_ssn	Mgr_start_date	Dnumber	Dlocation	Pname	Pnumber	Plocation	Dnum	Essn	Pno	Hours	Essn	Dependent_name	Sex	Bdate	Relationship	L3	CO3	
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	d) Write a RA expression to list the names of all employees with two or more dependents. e) Write a RA expression to retrieve the names of employees who have no dependents. f) Write a RA expression to list the names of managers who have at least one dependent. g) Write a RA expression to make a list of project numbers for projects that involve an employee whose last name is 'smith', either as worker or as a manager of the department that controls the project. h) Write a RA expression to retrieve the first name, last name and salary of all employees who work in the department 5. i) Write a RA expression to select the tuples for all employees who either work in the department 4 and make over \$25000 per year or work in the department 5 and make over \$30000.																																										
2	<table border="1"> <thead> <tr> <th>ID</th><th>Name</th><th>Dept_Name</th><th>Salary</th></tr> </thead> <tbody> <tr> <td>10101</td><td>John</td><td>Biology</td><td>65k</td></tr> <tr> <td>12121</td><td>Robin</td><td>Computer science</td><td>90k</td></tr> <tr> <td>25252</td><td>Alya</td><td>Electrical</td><td>40k</td></tr> <tr> <td>26589</td><td>Yusuf</td><td>Finance</td><td>95k</td></tr> <tr> <td>54789</td><td>Ravi</td><td>Music</td><td>60k</td></tr> <tr> <td>78787</td><td>Raj</td><td>Physics</td><td>87k</td></tr> <tr> <td>87458</td><td>Jayant</td><td>History</td><td>75k</td></tr> <tr> <td>76985</td><td>Pratik</td><td>Computer science</td><td>89k</td></tr> <tr> <td>12547</td><td>Neil</td><td>Finance</td><td>80k</td></tr> </tbody> </table> a) Write a RA expression to find all the instructors working in "Music" department. b) Write a RA expression to find all the instructors drawing salary between 50k to 80k c) Write a RA expression to find all instructors with salary greater than 75K. d) Write a RA expression to find all instructors who are working in "physics" department and drawing salary greater than 90k. e) Write a RA expression to find the name of all instructors who belong to Electrical and computer science department. f) Write a RA expression to list all instructors ID, name, and salary but donot care about the dept_name.	ID	Name	Dept_Name	Salary	10101	John	Biology	65k	12121	Robin	Computer science	90k	25252	Alya	Electrical	40k	26589	Yusuf	Finance	95k	54789	Ravi	Music	60k	78787	Raj	Physics	87k	87458	Jayant	History	75k	76985	Pratik	Computer science	89k	12547	Neil	Finance	80k	L3	CO3
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3	Explain the following RA concepts with examples: a) Select operation b) Project operation c) Unary operation d) Intersection operation e) MINUS operation f) Cartesian product g) Renaming h) Theta and Equi join i) Natural Join j) Full outer join k) Division operation l) Aggregate functions (SUM, COUNT, AVERAGE, MAXIMUM, and MINIMUM)	L2	CO3																																								

4	What are Join? List out and classify different types of joins. Explain each join with examples.	L2	CO3
5	How are assertions and triggers defined in SQL? Explain with examples.	L2	CO3
6	How to create and drop views in SQL? Explain with an example.	L2	CO3
7	Illustrate aggregate functions in SQL. Write a query to find the sum of the salaries of all employees, the maximum salary the minimum salary and the average salary by renaming the columns in a single row table.	L3	CO3
8	Explain EXISTS and UNIQUE functions in SQL. Consider the company database given in Question number 1, write a query to list the name of the manager who have atleast one dependent.	L3	CO3
9	Explain an informal design guideline for relational schema design.	L2	CO4
10	What is the need for normalization? Explain 1NF, 2NF and 3NF with examples.	L2	CO4
11	What is functional dependency? Explain the inference rules for functional dependency with proof. Also explain types of anomalies with examples.	L2	CO4
12	What do you mean by normalization? Explain BCNF and 2NF with a suitable examples.	L2	CO4
13	Consider the following schema: Sailors (Sid, Sname, rating, age) Boats (bid, bname,color) Reserves (Sid, bid,day) Write queries in SQL. a) Find the names of sailors who have reserved at least one boat. b) Find sailors whose rating is better than some sailors called 'Rocky'. (Use nested query) c) Find the average age of sailor for each rating level that at least two sailors. d) Find the name and age of the oldest sailor.	L3	CO3
14	Discuss the ACID properties of database transaction, and also explain transition diagram of transaction.	L2	CO5
15	Why concurrency controls is needed? Demonstrate with an example.	L2	CO5
16	Discuss two-phase locking technique for concurrency control.	L2	CO5
17	When deadlock and starvation problem occur? Explain how these problems can be resolved.	L2	CO5
18	a) Explain transaction support in SQL. b) Explain different types of locks used in concurrency control.	L2	CO5
19	What is NOSQL? Explain CAP theorem.	L2	CO5
20	What are document based NOSQL systems? Explain basic CRUD operations in MongoDB.	L2	CO5

Sign of the faculty
Darshan A

HOD